Fiscal Year:	FY 2022	**Task Last Updated:**	FY 08/03/2021
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Space Flight Environment Induces Remodeling of Vascular Network and Glia-Vascular Communication in Mouse Retina		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS Rodent Research-9 NOTE: End date changed to 10/31/2022 per NSSC (Ed., 12/14/21) NOTE: End date changed to 10/31/2021 per F. Hernandez/ARC (Ed., 11/6/18) NOTE: End date changed to 3/31/2020 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 4/30/2019 per F. Hernandez/ARC (Ed., 6/27/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University) Sun, Shu-Wei Ph.D. (Loma Linda University)		
Grant/Contract No.:	NNX15AB41G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The eye is a unique organ because it is relatively unprotected and constantly exposed to atmospheric oxygen, background radiation, environmental chemicals, and physical abrasion. A recent report shows that more than 30% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. Our Preliminary studies from mice that had been subjected to space flights showed that environmental conditions during space travel lead to oxidative stress and induce adverse microvessel remodeling in the retina. However, our knowledge about the pathophysiological process from adaptive response to irreversible oxidative damage in the retina during and after space flight exposure and the underlying cellular mechanism(s) of these disturbances is very limited and inconsistent. The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against space flight environment-induced changes in retinal neurovascular coupling and retinal function. The hypothesis will be tested in the specific aims with the space flight experiments on the International Space Station (ISS) and compared to that of ground-based groups using matured female mice. For flight studies, there will be two experiments over two flights, one for in-flight adaptation study on the ISS and one for studying re-adaptation after live animal return to Earth following long-term ISS mission. Selected groups of mice will be treated intraperitoneally with an antioxidant a week before launch and weekly during space flight. Animals will be sacrificed in orbit at 14 and 30 days after launch or 30 days and up to 6 months after return alive on Earth. Together, our unique, integrative, and quantitative activities with advanced MRI (magnetic resonance imaging) imaging techniques and retinal functional testing using visual electroretinography will provide insight into the molecular mechanisms and pathways of space flight condition-induced mitochondrial oxidative damage on retinal BRB integrity, vascular remodeling, and retinal function. The data will provide unique criteria for risks of functional detriments associated with space flight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our proposed study from space flight mission to ISS will provide a unique opportunity to evaluate the effects of space hazards and mechanisms of response and remodeling of the ocular system. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Task Progress:	The health risk from space flight-induced neuronal damage and potential adverse neurovascular effects is a chief concern. More recently, it has been proposed that neuroinflammatory response plays an important role in the neurovascular remodeling in the brain after stress. One of the goal of the present study was to characterize the changes in the gene expression of neuroinflammation panel for inflammation, neuronal function, metabolism, and stress in mouse brain tissue and to quantify circulating cytokines in the plasma after space flight. Ten-week old male C57BL/6 mice were launched to the International Space Station (ISS) on SpaceX-12 for a 35-day mission, Within 38+4 hours of splashdown, mice were returned to Earth alive. Brain tissues and whole blood were collected for analysis. Habitat ground control (GC) mice were maintained on Earth in flight hardware cages. A novel digital color-coded barcode counting technology (NanoStringTM) was used to evaluate gene expression profiles in the space flight mouse brain. The Neuroinflammation panel includes 757 genes covering the core pathways and processes that define neuroimmune interactions. A set of 52 differently expressed genes significantly ($p<0.05$) segregates the GC group from the flight (FLT) group. Many pathways associated with cellular stress, inflammation, apoptosis, and metabolism were significantly altered by flight condition. Genes supporting neuronal synaptic signaling and migration were significantly downregulated in FLT compared to the GC mice. A decrease in the expression of genes important for oligodendrocyte differentiation and myelin sheath maintenance was also observed. Moreover, mRNA expression of many genes related to antiviral signaling, reactive oxygen species (ROS) generation, and bacterial immune response were significantly downregulated. Changes in circulating cytokine/chemokine expression were also measured in the plasma using Luminex assay. Levels of several cytokines/chemokines in the flight group were significantly altered ($p<0.05$) in the plasma compared to GC controls, including IL-12, IP-10, MCP-1 MIP-1a, and MIP-2. These data indicate that neuroinflammation and altered immune reactions may be closely associate with space flight–induced stress response and have an impact on the neurovascular function that may result in neurodegeneration. We also have characterized the effects of space flight on oxidative damage in the mouse brain and its impact on blood-brain barrier (BBB) integrity. Quantitative assessment of brain tissue demonstrated that space flight caused an up to 2.2-fold increase in apoptosis in the hippocampus compared to the control groups. Immunohistochemical analysis of the mouse brain revealed a strong increase in expression of aquaporin-4 (AQP-4) in the flight hippocampus compared to controls. There was also a significant increase in the expression of platelet endothelial cell adhesion molecule-1 (PECAM-1) and a decrease in the expression of the BBB-related tight junction protein, Zonula occludens-1 (ZO-1). These results gave strong indication of a disturbance of the BBB integrity. Quantitative proteomic analysis showed significant alterations in pathways responsible for mitochondrial function, neuronal structure, neurovascular integrity, protein/organelle transport and metabolism in the brain after space flight. Changes in pathways associated with adhesion and molecular remodeling were also documented. These data indicate that long-term space flight may have pathological and functional consequences associated with neurovascular damage and late neurodegeneration.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Mao XW, Stanbouly S, Nishiyama N, Delp M. "Spaceflight induces mouse retina changes near optic nerve head." 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020

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